

有理数类

C++

```
1  #include <iostream>
2  #include <algorithm> // 包含算法库，用于求最大公约数
3
4  using namespace std;
5
6  class Rational {
7  private:
8      int numerator;    // 分子
9      int denominator; // 分母
10
11 public:
12     // 默认构造函数，带默认值，简化分数形式
13     Rational(int num = 0, int denom = 1) {
14         int gcd = __gcd(num, denom); // 使用 __gcd 函数求最大公约数
15         numerator = num / gcd;
16         denominator = denom / gcd;
17     }
18
19     // 加法
20     void add(const Rational &r1, const Rational &r2) {
21         numerator = r1.numerator * r2.denominator + r2.numerator * r1.denominator;
22         denominator = r1.denominator * r2.denominator;
23         simplify();
24     }
25
26     // 减法
27     void minus(const Rational &r1, const Rational &r2) {
28         numerator = r1.numerator * r2.denominator - r2.numerator * r1.denominator;
29         denominator = r1.denominator * r2.denominator;
30         simplify();
31     }
32
33     // 乘法
34     void multi(const Rational &r1, const Rational &r2) {
35         numerator = r1.numerator * r2.numerator;
36         denominator = r1.denominator * r2.denominator;
37         simplify();
38     }
39
40     // 除法
41     void divide(const Rational &r1, const Rational &r2) {
42         numerator = r1.numerator * r2.denominator;
43         denominator = r1.denominator * r2.numerator;
44         simplify();
45     }
46
47     // 打印分数形式
48     void printFormal() const {
49         cout << numerator << "/" << denominator;
50     }
51
52     // 打印小数形式，限制小数点后两位
53     void printDecimal() const {
54         double result = static_cast<double>(numerator) / denominator;
55         printf("%.2f", result); // 使用 printf 控制小数位数
56     }
57
58     // 打印分数和小数形式
59     void printBoth() const {
```

```

60     cout << "Formal format: ";
61     printFormal();
62     cout << ", Decimal format: ";
63     printDecimal();
64     cout << endl;
65 }
66
67 private:
68     // 简化分数
69     void simplify() {
70         int gcd = __gcd(numerator, denominator); // 使用 __gcd 函数求最大公约数
71         numerator /= gcd;
72         denominator /= gcd;
73     }
74 };
75
76 //StudybarCommentBegin
77 int main()
78 {
79     int firstN, firstD, secondN, secondD;
80     char op;
81
82     // cout<<"Please input the numerator of first Rational: ";
83     cin>>firstN;
84     // cout<<"Please input the denominator of first Rational: ";
85     cin>>firstD;
86     // cout<<"Please input the numerator of second Rational: ";
87     cin>>secondN;
88     // cout<<"Please input the denominator of second Rational: ";
89     cin>>secondD;
90
91
92     Rational r1(firstN, firstD), r2(secondN, secondD), r3;
93
94
95     cin>>op;
96
97     cout<<"the Formal format of the first rational is : ";
98     r1.printFormal();
99     cout<<"the Formal format of the second rational is : ";
100    r2.printFormal();
101    cout<<endl;
102
103    switch(op)
104    {
105        case '+':
106            r3.add(r1, r2);
107            r3.printBoth();
108            break;
109        case '-':
110            r3.minus(r1, r2);
111            r3.printBoth();
112            break;
113        case '*':
114            r3.multi(r1, r2);
115            r3.printBoth();
116            break;
117        case '/':
118            r3.divide(r1, r2);
119            r3.printBoth();
120            break;
121        default:
122            cout<<"Invalid operator!";
123    }
124
125    return 0;
126 }
127 //StudybarCommentEnd

```